# A Discrete Multitone Wireline Transceiver With Clipping Ratio Optimization For ADC-Based High-Speed Serial Links

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***Abstract*—In this paper, we investigated the relationship between clipping ratio and bit-error rate (BER) for a discrete multitone (DMT) wireline transceiver. The peak-to-average-power ratio (PAPR) controls the trade-off between signal-to-noise ratio (SNR) and non-linear distortion, such as clipping. The bit-error-rate (BER) performance degrades for both cases: high distortion due to the low PAPR and low SNR caused by the high PAPR. The optimal spot is obtained by conducting a sweep simulation on MATLAB software. The simulation results demonstrate that the 1E-6 order BER is achieved with the 160 Gb/s data rate and 12.3 dB PAPR when communicating over a channel, exhibiting 18 dB channel insertion loss (IL) at Nyquist frequency.**

***Index Terms*—ADC-based serial link, wireline transceiver, equalization, pulse amplitude modulation, PAM, discrete multitone, DMT, power loading, peak-to-average-power ratio, PAPR.**

## I. Introduction

With growing data traffic that is encountering high-performance computing, the requirement for higher communication bandwidth increases, driving the development of 4-level pulse amplitude modulation (PAM-4) with enhanced spectral efficiency. Thanks to the recent advancement of a time-interleaved analog-to-digital converter (TI-ADC) and the scaling of CMOS technology, ADC-based wireline transceiver (TRX) [1], [2] have shown strong equalization capability and high reconfigurability. Although PAM-4 has better power spectral efficiency than non-return-to-zero (NRZ) signaling, design complexity increases significantly.

Recently, discrete multitone (DMT) modulation [3]–[5] has paid great attention due to its feedforward datapath structure with high energy efficiency. When cyclic-prefix (CP) is suitably long enough to cover the intersymbol interference (ISI), frequency domain equalizer (FDE) with a single tap complex multiplier is enough to cancel the dispersion. Because data converters and analog front ends (AFEs) have a finite dynamic range, non-linear distortion, such as clipping, is inevitable at the TX side. Such an aggressive clipping results in performance degradation, while the overall signal signal-to-noise ratio (SNR) increases when proper clipping is applied. In this paper, we investigated the relationship between clipping ratio and performance with the MATLAB simulation. We found that increasing power results in sub-optimal performance due to the severe clipping effect.

## II. Clipping of DMT

The DMT signals often show significantly higher swings than average power, making data converters lead to high distortions, such as clipping. As the law of large numbers (LLN), this time-domain DMT samples follow the Gaussian when the quadrature amplitude modulation (QAM) symbols are selected from a uniform distribution. Then, the clipping probability is represented by the peak-to-average-power ratio (PAPR) as follows.

$$Pr(|x| \geq M) = 2 \cdot erfc(\frac{M}{x_{rms}}) = 2 \cdot erfc(\text{PAPR}) \quad (1)$$

where $x$ is the time-domain DMT symbols, $x_{rms}$ is the root-mean-square (RMS) power of $x$, $M$ is the maximum range of the data converters, and $erfc(\cdot)$ denotes the complementary error function. Due to the orthogonality between subtones in DMT, the RMS power is the sum of power loading as,

$$x_{rms}^2 = \sum_{k=1}^{N} ||P_k||^2 \quad (2)$$

where $P_k$ is the power loading for the $k$-th subchannel, and $N$ is the size of fast Fourier transform (FFT). By comparing Eq. 1 and Eq. 2, we notice that enhancing SNR by increasing power loading leads to high clipping probability and degrading bit-error-rate (BER) performance.

## III. Simulation Results

Fig. 1 shows the overall block diagram of the DMT TRX datapath. In the transmitter (TX) side, the power loader pre-emphasizes each constellation from a bit loader according to the given SNR profile. Then, the inverse FFT (IFFT) processor transforms signals from the frequency domain to the time domain before inserting CP. The IFFT engine is implemented in a partially-serial manner to minize silicon area while keeping communication throughput. The receiver (RX) is implemented with the CP removal, FFT processor, and FDE. The FDE magnifies and rotates the constellation of FFT output based on the channel estimation (CHEST) results. The CHEST is implemented with the sign-sign least mean square (SS-LMS) to adapt the channel variation with minimal power consumption and silicon area overhead.

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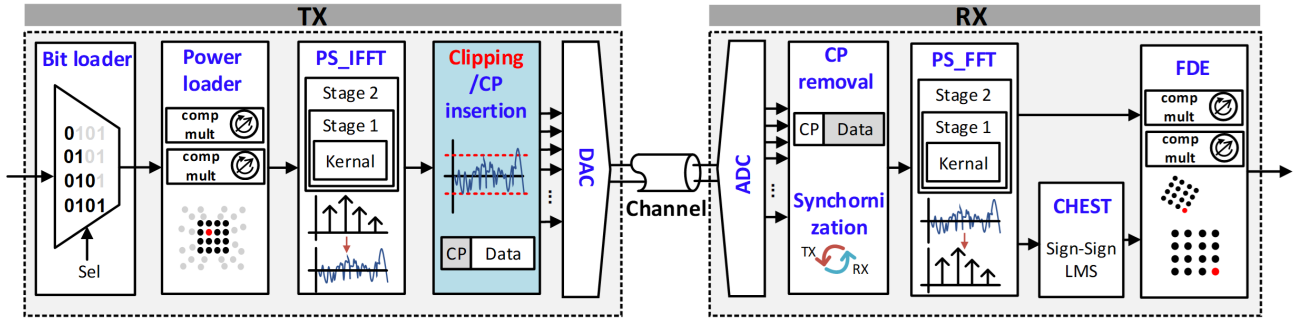
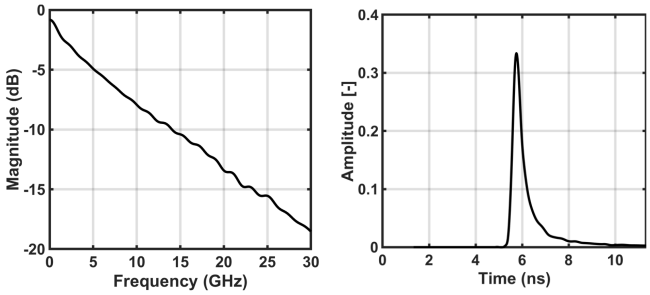



Fig. 1: Overall DMT TRX structure.




(a) (b)

Fig. 2: Simulated channel pulse response in frequency-domain (a), and time-domain (b).

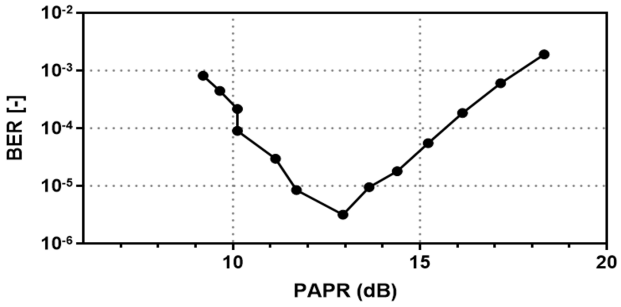



Fig. 3: Simulated PAPR versus BER curve under fixed data rate.

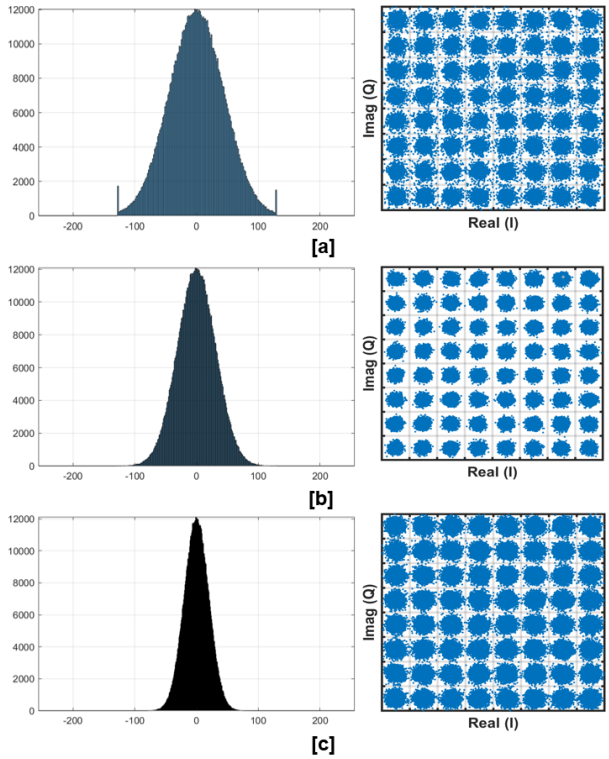



Fig. 4: Histogram of the clipped DMT symbol (left), and constellation diagram after RX FDE (right) with aggressive clipping (a), optimal clipping (b), without clipping (c).

Fig.2 shows the channel response in frequency-domain (a) and time-domain (b). The channel exhibits 18 dB channel loss at 30 GHz Nyquist frequency. The target data rate is 180 Gb/s, achieving 6 b/Hz spectral efficiency.

Fig. 3 shows the relationship between PAPR and simulated BER. For the low PAPR region, the BER performance improves as PAPR increases since the overall SNR is enhanced. In contrast, the BER rather increases since the distortion caused by aggressive clipping becomes dominant. The optimal BER is about 1E-6 when the PAPR is 12.3 dB. Fig.4 shows the constellation and histogram depending on the PAPR. For both low and high PAPR conditions, the BER performance is sub-optimal because of the low SNR or the high distortion.

## IV. Conclusion

In this paper, we investigated the optimal clipping ratio with the MATLAB simulation. The experimental results show that the optimal PAPR is 12.3 dB, achieving 1E-6 BER.

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